AMD Developer Hackathon / Track 1

Agentic Reproducibility Engine

- AMD-hosted multi-agent research reproducibility evaluator.
- Powered by Qwen/Qwen3.5-27B, vLLM, ROCm, and Hugging Face Spaces.

Reproducibility evidence is scattered

Problem

- Claims, code, data, and methods live across disconnected systems.
- Manual audits are slow, inconsistent, and often hide uncertainty.

Visible multi-agent audit workflow

Solution

- Submit a paper and choose autonomy.
- Watch the trace, tool calls, evidence, verifier objections, and final report.

Separate agents, separate artifacts

Agentic Workflow

- Plan audit scope.
- Read claims and artifacts.
- Resolve external evidence.
- Score, plan, verify, and report.

Static frontend + AMD-hosted backend

AMD Architecture

- Hugging Face Static Space hosts index.html.
- FastAPI agent backend runs on AMD Developer Cloud.
- vLLM serves Qwen on ROCm through an OpenAI-compatible endpoint.

Live resolvers, fail-closed decisions

Evidence Tools

- arXiv, DOI, GitHub, and dataset resolver tools.
- Missing evidence becomes an explicit gap.
- The verifier can degrade the decision.

Judges inspect the work

Demo Flow

- Submit paper -> autonomy slider -> live trace.
- Evidence panel -> verifier objections -> final report.

Agent behavior is tested

Evaluation

- Prompts, contexts, artifacts, and tool calls are graded.
- Resolver coverage and verifier behavior are checked.
- Placeholder evidence is explicitly banned.

Faster first-pass reproducibility audits

Business Value

- For labs, ML teams, funders, reviewers, and benchmark maintainers.
- Outputs provenance, uncertainty, and concrete replication next steps.

From demo to research infrastructure

Roadmap

- Deploy AMD/Qwen backend and publish HF Space.
- Expand resolver sources and eval cases.
- Add PDF ingestion and team workflows.